



Lecture 1

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Introduction

kNN (k-nearest neighbours)

Tagpfauenauge (*Aglais io*)

- ♦ Das Tagpfauenauge wurde zum Schmetterling des Jahres 2009 gewählt und ist einer der häufigsten Schmetterlinge in Deutschland, Österreich und der Schweiz.
- ♦ Das unverkennbare und auffälligste Merkmal sind die an jeder Vorder- und Hinterflügelspitze gut erkennbaren, schwarz, blau und gelb gefärbten Augenflecken. Die Falter erreichen eine Flügelspannweite von 50 bis 55 Millimetern. Sie haben eine rostrote Flügelgrundfärbung.
- ♦ Die Raupen werden circa 42 Millimeter lang. Sie sind schwarz am gesamten Körper, weisen zahlreiche feine weiße Punkte und schwarze Dornen auf und sitzen fast ausschließlich auf Brennnesseln.
- ♦ Auffällig ist die Abwehrstrategie des Falters: Nähert sich ein Angreifer, schlägt das Tagpfauenauge schnell mit den Flügeln, um die Augen darauf zur Geltung zu bringen und gibt dazu ein lautes Zischen von sich.

Quellen:

* <https://de.wikipedia.org/wiki/Tagpfauenauge>

* <https://www.plantura.garden/gruenes-leben/schmetterlinge-die-10-beliebtesten-heimischen-schmetterlingsarten>



Introduction and k NN

1. What is Machine Learning (ML)?
2. Some ML Applications
3. Kinds of Machine Learning
4. Supervised Learning: Classification and Regression
5. k -nearest neighbour (k NN) Classification and Regression



Your Turn !

Exercise 1

What is Machine Learning



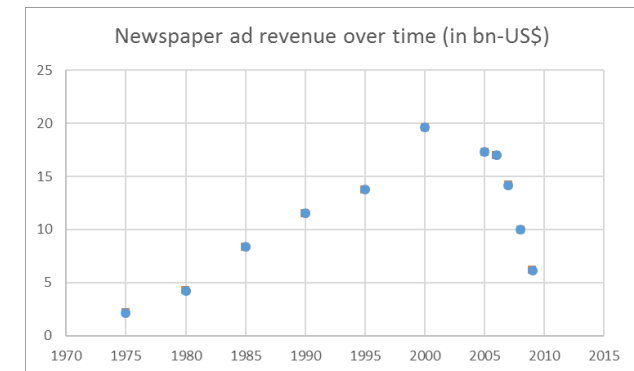


What is Machine Learning?

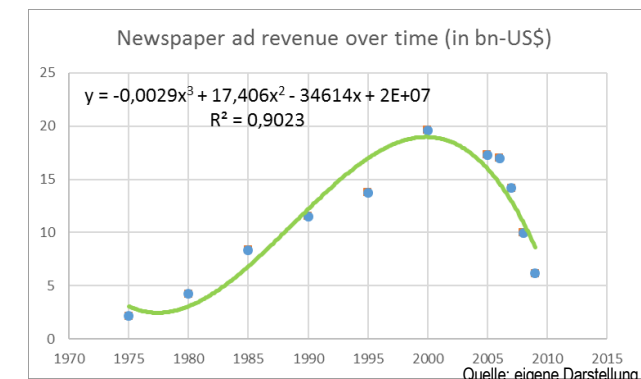
- ◆ Computer solves a problem without an explicitly coded algorithm
- ◆ Instead: use a highly parameterized algorithm and “learn” from data
- ◆ Model = Set of parameter values
- ◆ Compute the parameter values (the model) using a training algorithm and (lots of) examples (data)
- ◆ Most ML methods employ 2 algorithms
 - one for training the model and
 - one for model execution/evaluation

model inference

generally, we want less parameters than data points (re: generalisation) for the model to be actually useful bc if the model is tailor made exactly then it is most likely not applicable for usecases outside of those few data points



original dataset



polynomial regression of degree 3



Why use Machine Learning?

- ◆ For many real-world tasks, it is **difficult to come up with an explicit** (i.e. rule-based) **algorithm**
 - Too many rules needed to cover all the cases
 - Many exceptions (including exceptions to exceptions!) to be handled
 - Expert knowledge needed (i.e., an expert to handcraft the rules)
 - Rules can be difficult to
 - Design – rules interact in unpredictable ways
 - Maintain – adding new rules can easily break everything
 - Adopt to new domains – we need to significantly modify/add rules
 - Many tasks are often inherently subjective, it is difficult to model subjectivity with rules
- ◆ It is often **easier to manually label data** than to design an explicit algorithm
 - Labelling typically requires less expert knowledge
 - We don't care how complex or subjective the task is
 - We let the data „speak for itself” and machine learning algorithm to do the work
 - Labelled data may already be readily available



Photo by Austin Distel on Unsplash



Machine Learning is hard...



IN CS, IT CAN BE HARD TO EXPLAIN
THE DIFFERENCE BETWEEN THE EASY
AND THE VIRTUALLY IMPOSSIBLE.

© xkcd, <https://xkcd.com/1425/>, CC BY-NC 2.5

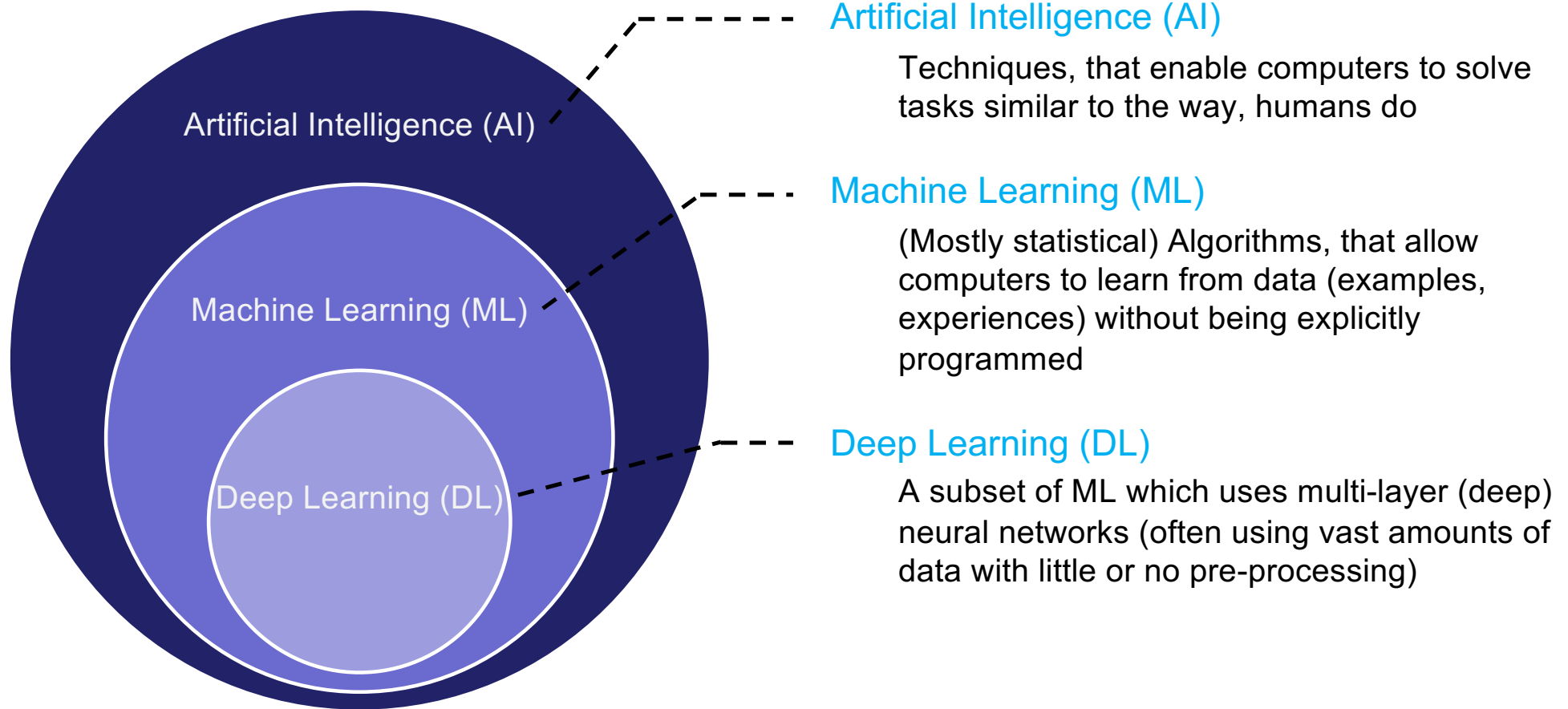


Chihuahua or Muffin?

© Twitter / Karen Zack / @teenybiscuit,
<https://twitter.com/teenybiscuit/status/707727863571582978>



AI vs. ML vs. DL



Weak AI



vs.

artificial general intelligence

Strong AI (AGI)



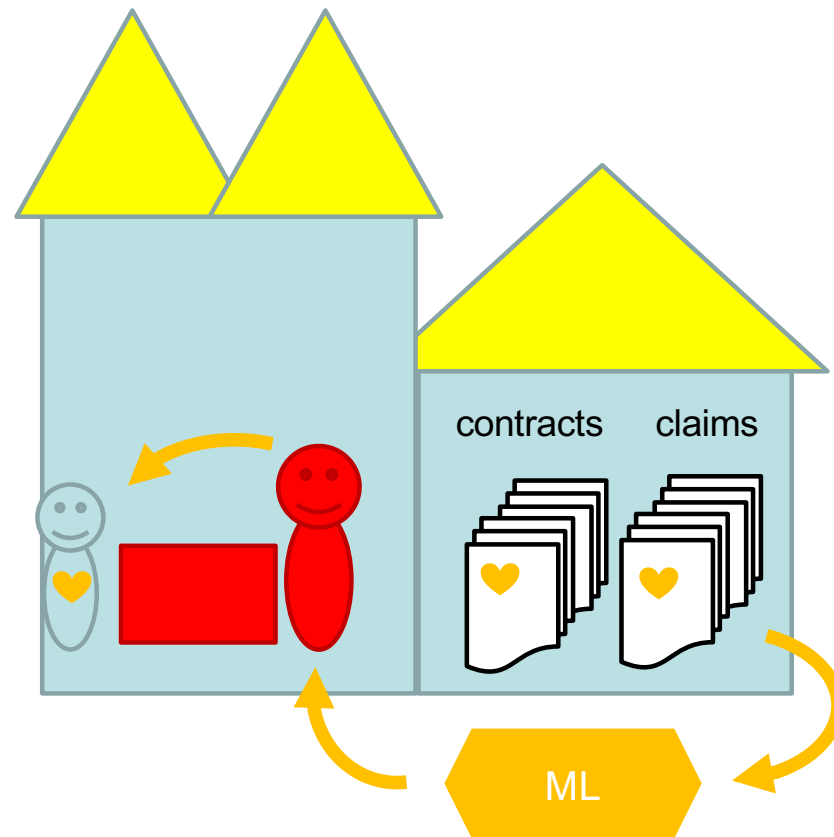


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Insurance Risk





Industrial Applications: quality assurance in fruit and vegetable analysis



From: Dubey, Shiv Ram, and Anand Singh Jalal. "Application of image processing in fruit and vegetable analysis: A review." *Journal of Intelligent Systems* 24(4): 405-424, 2015.



License-Plate Recognition

- electronic toll collection,
- law enforcement,
- speed-limit enforcement



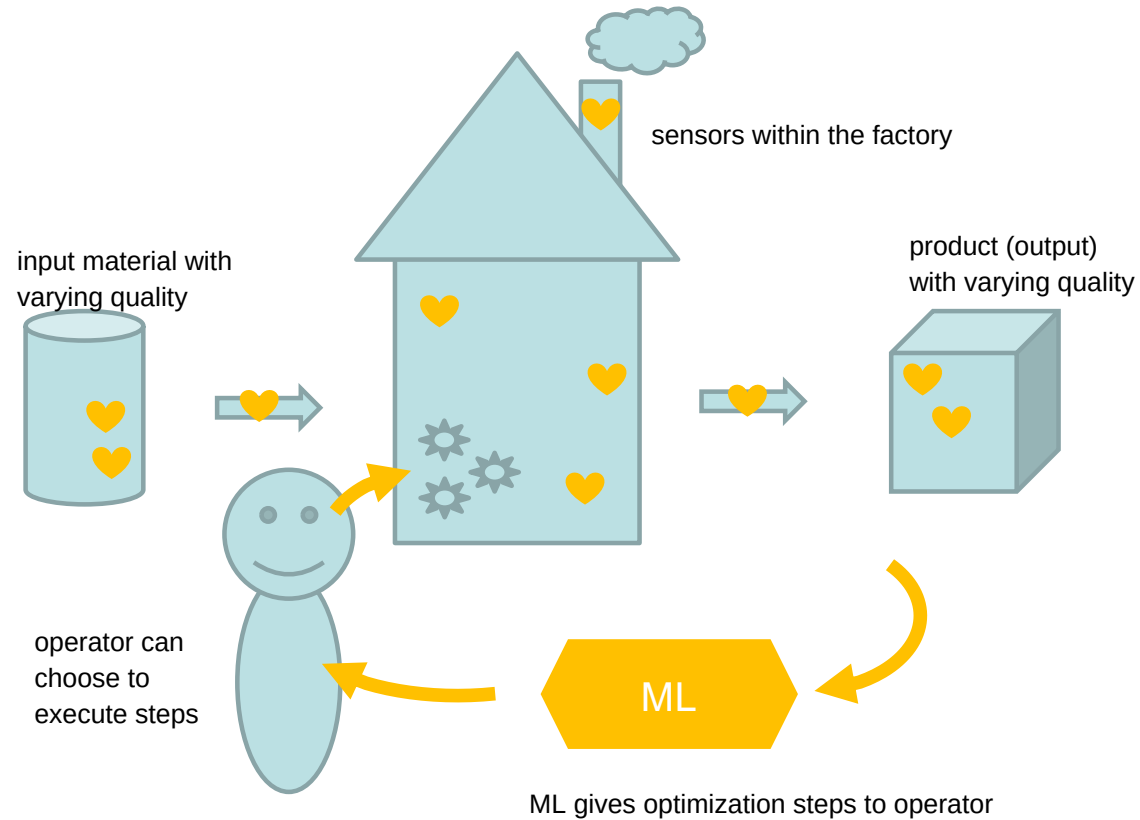
Brenner toll station, © Peter Müller, CC BY-SA 3.0 DE



© Apfel3748 / B. Seidl, CC BY-SA 3.0



Production Optimization





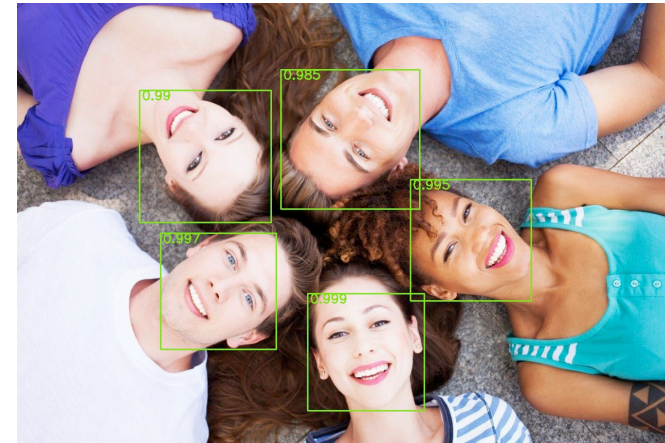
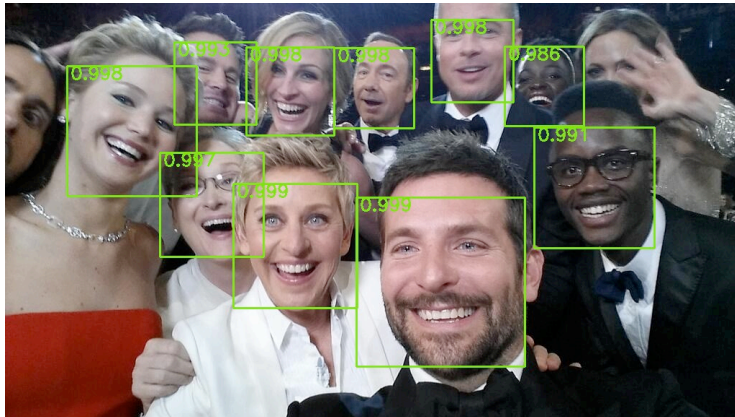
Speech Processing

- ◆ We can learn to transcribe speech (*speech to text*)
- ◆ We can learn to index speech (*keyword search*)
 - *It's hard to recognize speech.*
 - *It's hard to wreck a nice beach.*
- ◆ We can learn to understand speech
 - Command and control
 - Information extraction





Face Recognition in Digital Cameras/Smartphones



Farfadi, Sachin Sudhakar, Mohammad J. Saberian, and Li-Jia Li. *Multi-view face detection using deep convolutional neural networks*. Proc. 5th ACM International Conference on Multimedia Retrieval. ACM, 2015.



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Kinds of machine learning

◆ Supervised learning

- Labelled data points (class or value)
- Goal: study many labelled examples, make predictions about future data points
- **Tasks**: classification, regression, link prediction/recommender, ...

Semi-supervised: very few labelled data points, not enough to be supervised

◆ Unsupervised learning

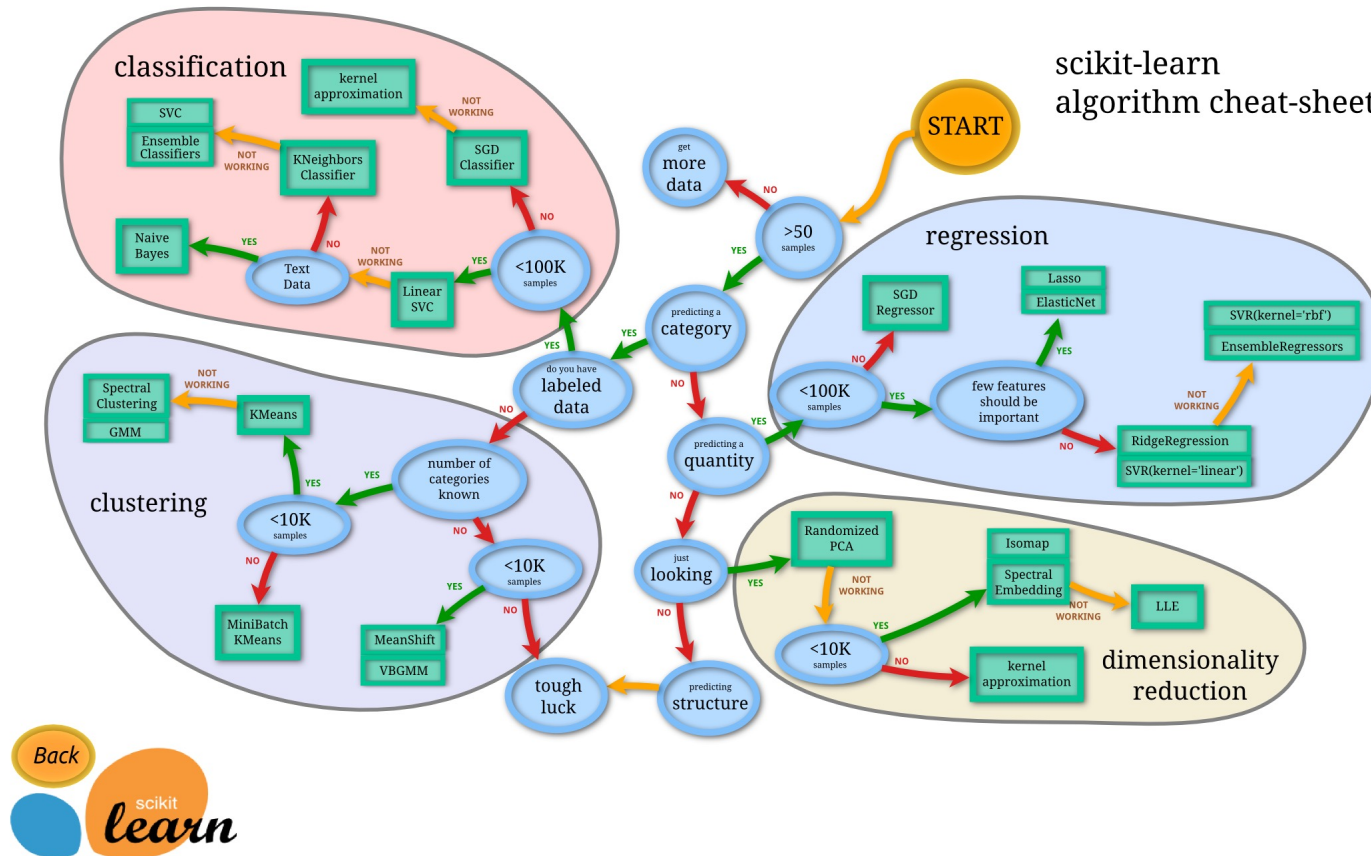
- Data points are not labelled
- Goal: organize the data in some way or describe its structure
- **Tasks**: clustering, anomaly detection, dimensionality reduction, association rules, topic-modelling, ...

◆ Reinforcement learning

- Choose an action in response to a data point, receive reward signal
- Goal: modifies the response strategy in order to achieve the highest reward

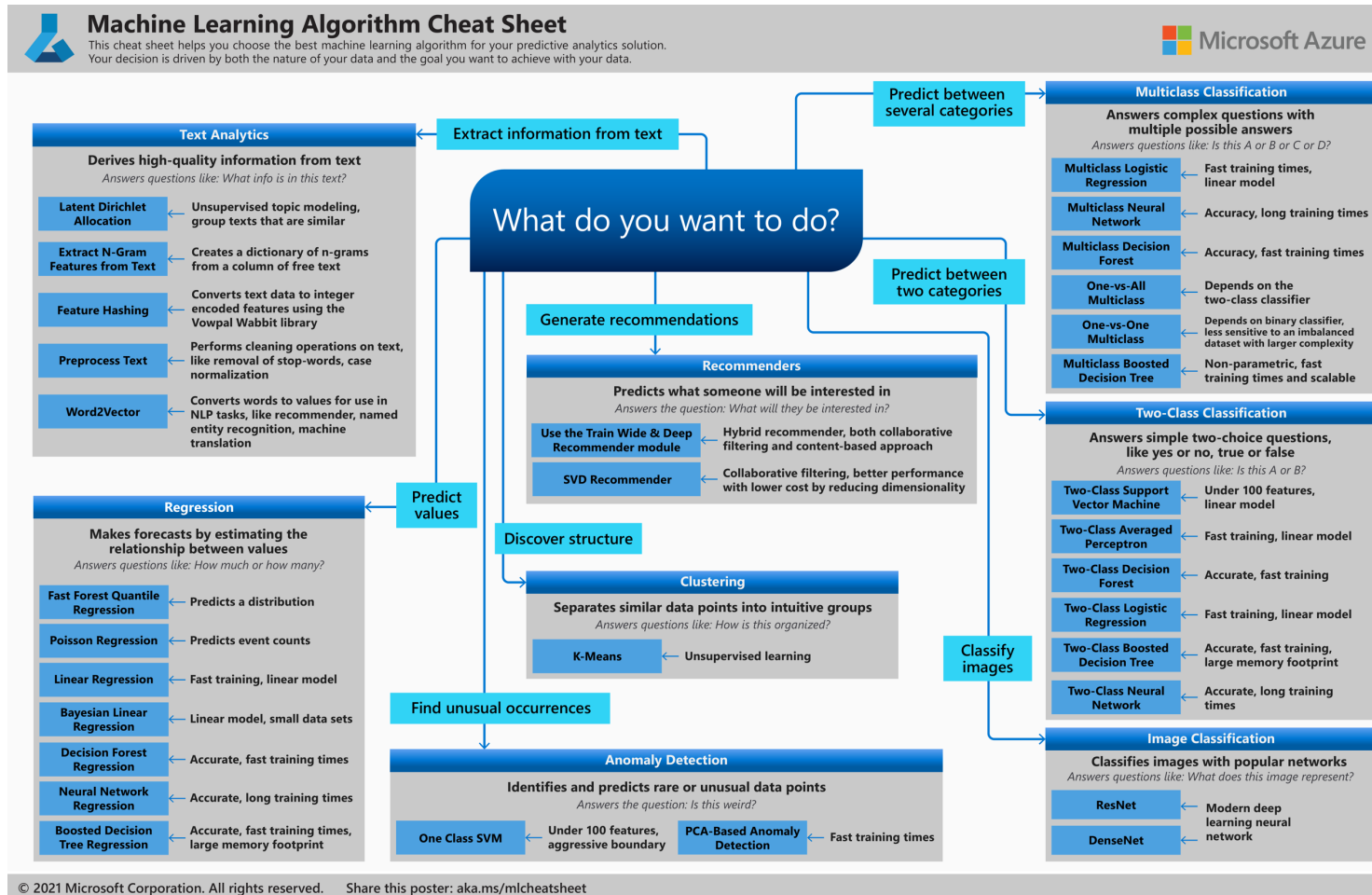


Machine Learning Tasks and Algorithms – scikit learn





Machine Learning Tasks and Algorithms - Microsoft



Source: <https://docs.microsoft.com/en-us/azure/machine-learning/algorithm-cheat-sheet>



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Supervised Learning

♦ Supervised learning

- Learn from labelled data points (class or value)

$$D = \{(x_i, y_i), i = 1, \dots, n\}$$

- with identical independent distribution (i.i.d.) assumption
- the x_i are n -dimensional feature vectors
- **Classification**: labels are categorical
- **Regression**: labels are numerical

♦ “Learn” the mapping between input values and output values

- single input x is called an **instance**, example, tuple or data point
- desired output y is called the **target** label (or just target)

♦ Classifier/Regressor f with parameters θ maps an instance x to a label y : $f(x|\theta) \rightarrow y$

- It may be allowed to **reject** certain x , if it cannot provide a reasonable y

♦ “Learning” (“Training”): optimize the parameters θ in order to maximize the prediction accuracy



Training a Classifier/Regressor

- ◆ Training: adjustment of model parameters θ so that a **measure of error** is minimized
 - **Training error**: error computed on the labelled training set error is defined
 - Minimization using an **optimization method** (e.g. OLS)
 - ML algorithms differ in optimization criteria and optimization method
- ◆ We want to know how classifier/regressor works on new, unseen instances
 - This property is called **generalization** – the classifier/regressor should generalize well
 - **Testing error**: error computed on (labelled) instances not used for training
- ◆ Frequently: maximize a **score** instead of minimizing an **error**



Classification

- ◆ **Classification:** labels are categorical
 - Most classifiers return a “probability” together with the class
- ◆ **Types of classifiers**
 - **Binary classification:** only two output labels (yes/no, 0/1, -1/+1, positive/negative)
 $y \in Y = \{-1, +1\}$ or $y \in Y = \{0, 1\}$
 - **Multi-class classification:** each instance has one of C labels
 $y \in Y = \{1, 2, 3, \dots, C\}$
 - **Multi-label classification:** an instance can have more than one label at once
- ◆ **It is often easy to turn**
 - Binary classification into regression
 - Regression into binary classification



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k Nearest Neighbor (k NN) Classifier

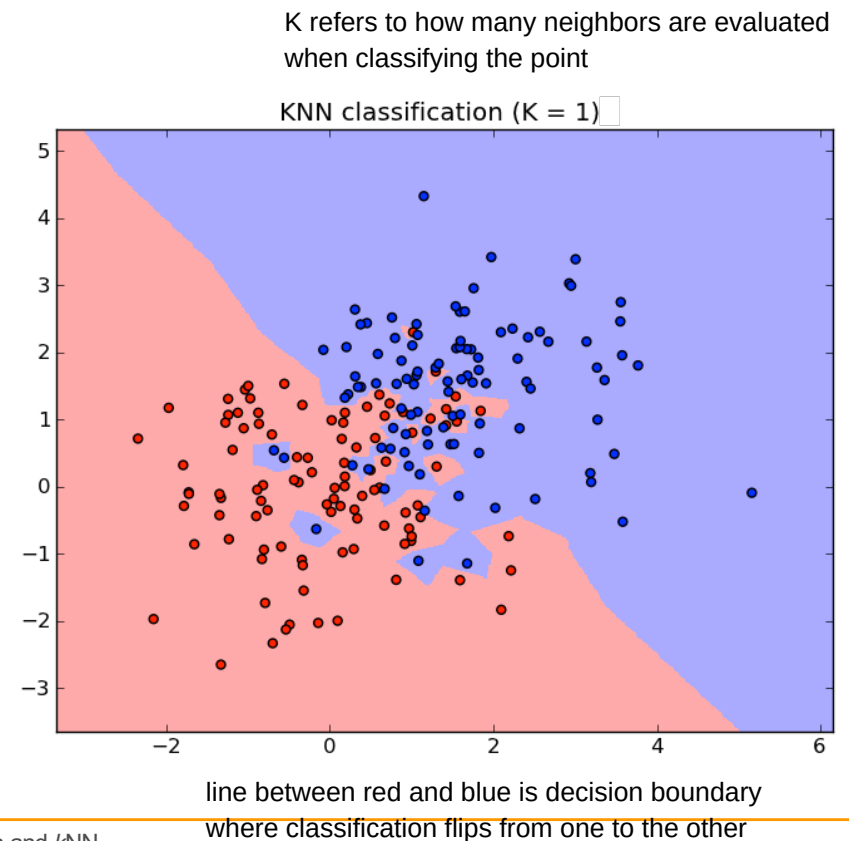
- ◆ “Lazy” learning, instance-based classification

- ◆ Earliest models date back to 1970s

- ◆ Select class based on nearest neighbor

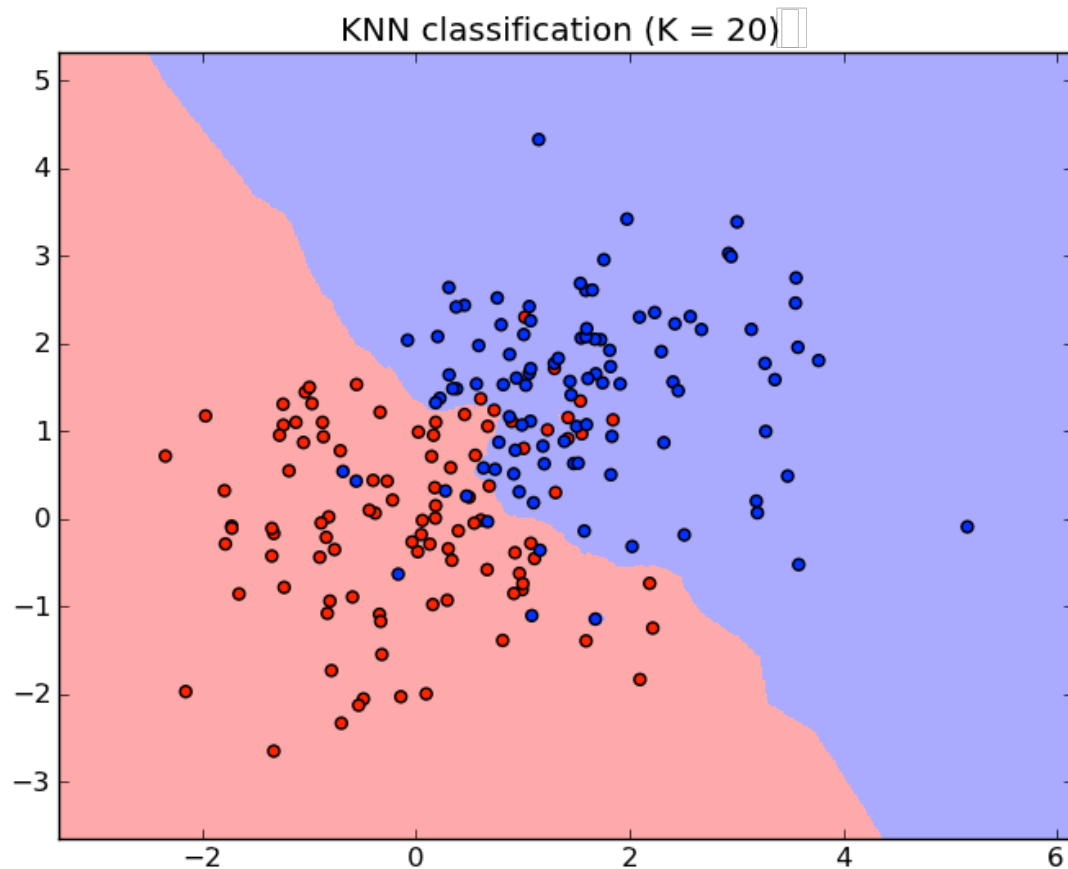
any point on the graph looks at its nearest neighbor and classifies itself accordingly

stark parametrisiert





Putting the k in kNN



the higher K is, the better the classification, counteracting errors (see blue data points in red)

in this example, 20 neighbors are evaluated before the point is classified

it is also possible to apply weight to the points based on e.g. distance

→ k acts as a smoother to the decision boundary



Probabilistic Interpretation of k NN

probability if given data point belongs to certain class

- ◆ Estimate the conditional probability $P(y|\mathbf{x})$
by counting the data points in class y in the neighborhood of $\mathbf{x}$:

not a proper mathematical probability and more
a percentage of how sure the model is of the correctness
of its classification

$$P(y|\mathbf{x}) = \frac{\#y}{k}$$

- ◆ **Effect of hyperparameter k :** variance-bias tradeoff

- Small k leads to large **variance** and unreliable classification
- Large k leads to large **bias** and inaccurate estimation

meaning each data point has a lot of influence on the classification
(cf. $k = 1$) -> high variance, low bias

bias refers to how well the model conforms to the data points, cf. $k = 20$ where it is possible that there are actually pockets of red/blue in the opposite
cannot adapt as well as a small k can -> low variance, high bias

- ◆ **Hyperparameters:** parameters, which specifies details of the learning process.

- Hyperparameters: not learned during training.
- Parameters of the model (“the model”): learned during training.

each ML method inherently has a certain bias, meaning that depending on the desired result
and use case a method must be selected that has a bias which fits the use case
ML method will always return a result -> selection of correct method is important



kNN Pseudocode

◆ Input

- Finite set D of points to be classified
- Finite set T of training points
- Function $c: T \rightarrow \{1, \dots, m\}$ to determine class of training point
- Function $d: (a, b) \rightarrow \mathbb{R}$ to determine distance of points
- Natural number k

◆ Begin

- Foreach x in D do
 - Let $U \leftarrow \{\}$
 - Foreach t in T :
 - $U \leftarrow U + (d(x,t), c(t))$
 - $U \leftarrow \text{sort}(U)[:k]$ *// sort and keep k smallest entries*
 - Let $r(x)$ be the class with the highest number of occurrences in U

◆ Issues/Bottlenecks?



Distance Metrics

- ◆ Sometimes also called *kernel* or *norm*

- ◆ Euclidean Distance (L_2 -Norm):

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2} \quad \text{distance in two equal dimensions}$$

- ◆ Manhattan-Distance (L_1 -Norm):

$$d(\mathbf{x}, \mathbf{y}) = |x_1 - y_1| + |x_2 - y_2| + \dots + |x_n - y_n| \quad \text{sum of x and y dimension}$$

- ◆ Maximum-Norm (L_∞ -Norm):

$$d(\mathbf{x}, \mathbf{y}) = \max(|x_1 - y_1|, |x_2 - y_2|, \dots, |x_n - y_n|) \quad \text{evaluates each distance separately}$$

- ◆ Minkowski-Distance (L_p -Norm):

$$d(\mathbf{x}, \mathbf{y}) = \sqrt[p]{|x_1 - y_1|^p + |x_2 - y_2|^p + \dots + |x_n - y_n|^p}$$

- ◆ Cosine-Distance https://en.wikipedia.org/wiki/Cosine_similarity
- ◆ Jaccard Index https://en.wikipedia.org/wiki/Jaccard_index
- ◆ Hamming-Distance https://en.wikipedia.org/wiki/Hamming_distance
- ◆ Mahalanobis-Distance https://en.wikipedia.org/wiki/Mahalanobis_distance



Your Turn !

Exercise 2

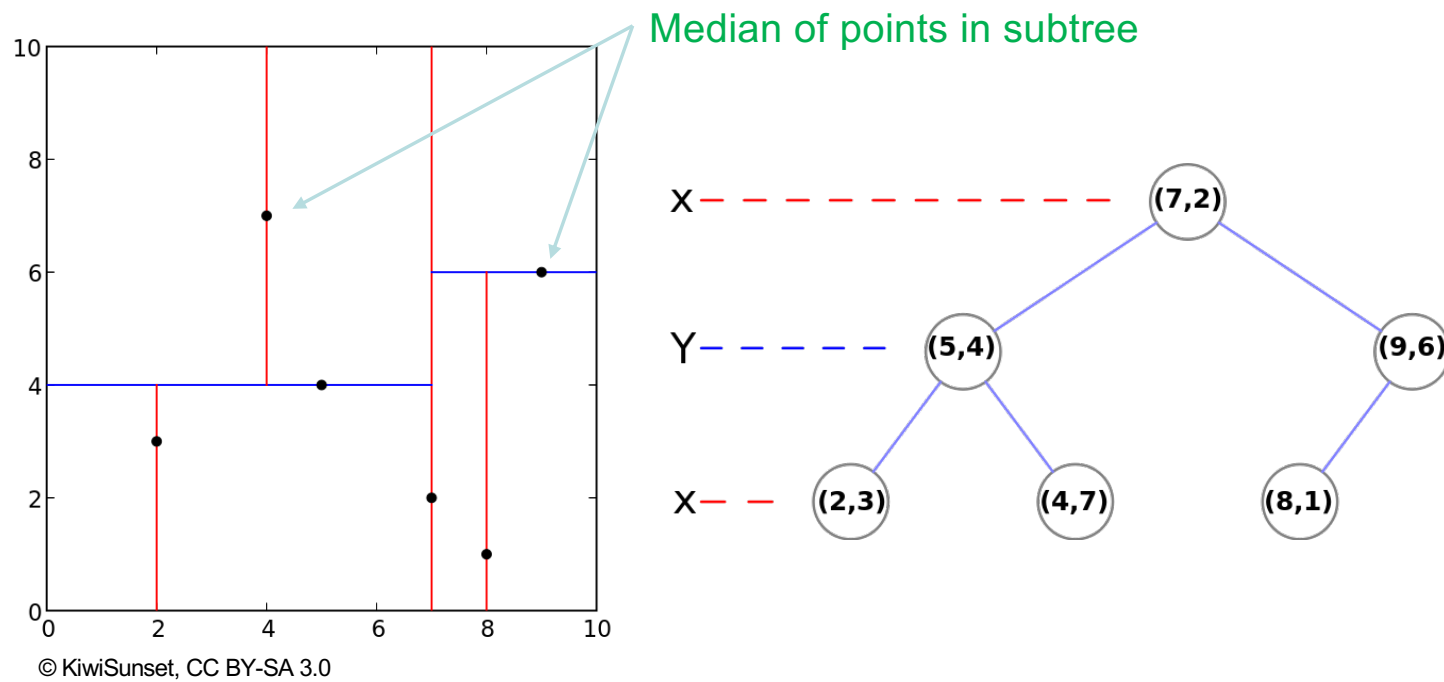
k NN





Optimize Data Storage

- ◆ Use a k-d tree
 - multi-dimensional Binary Search Tree (BST)
 - Maintain leaf counts on nodes





Questions

◆ How would you determine k ? hyperparameter tuning: try multiple values to find best one

◆ What about multi-class? majority vote, possibly weighted
needs large k and can quickly become complicated

◆ What about categorical data, or mixed data?

ordinal encoding (not ideal): each extra value is numbered starting from 0
- problem: orders values based on nothing and may skew probabilities/distances

generally, very difficult to incorporate and evaluate in a way that makes sense

independent ordinal encoding where each value is either 0 or 1 and each data
has a multi-vector that has a 1 for each applicable category



Curse of Dimensionality

- ◆ **Curse of Dimensionality**: there's a lot of space in high-dimensional data sets
 - Size of the data space grows exponentially with the number of dimensions
 - ➔ Size of your data set must also grow exponentially in order to keep the same density
 - ➔ If not, then data points start getting farther and farther apart.

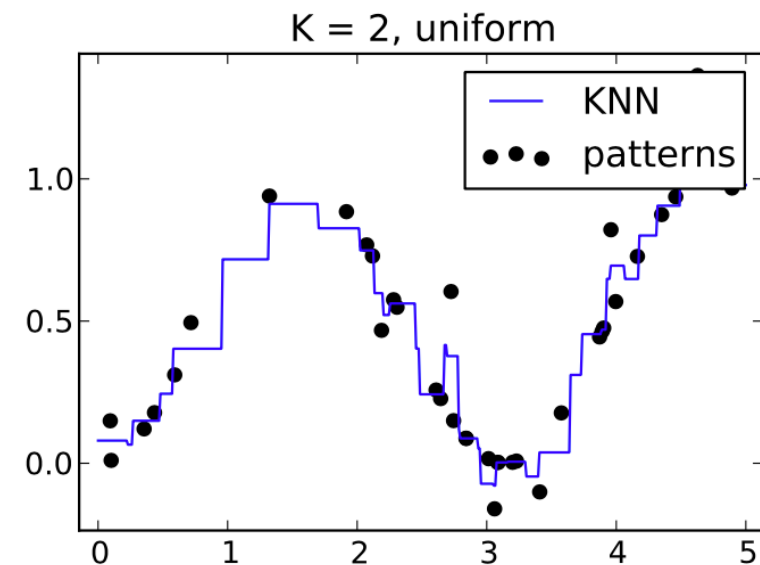
- ◆ Particularly bad for *k*NN
 - requires a data point to be close in every single dimension
 - each new dimension added, makes it harder for two specific points to be close to each
 - ➔ Ratio of closest distance to average distance increases (approaches 1)
 - ➔ If the closest point is almost as far away as the average point, then it has only slightly more predictive power than the average point.

- ◆ Solutions
 - Add (lots) more data (exponentially in the number of dimensions)
 - Reduce number of dimensions



kNN-Regression

- ◆ Target value is numeric (i.e., not classes)
- ◆ Select k closest samples
- ◆ Use an averaging function
- ◆ Can be useful with missing observations, if distance metric chosen carefully





Key Takeaways

- ◆ Machine Learning: Supervised, Unsupervised, Reinforcement
- ◆ Classification & Regression
- ◆ *kNN*: supervised and non-parametric
 - Able to learn highly non-linear class boundaries
 - Efficient implementation requires data organization/storage
 - Distance metric and dimensions need to be chosen carefully
 - Works great for small training sets
 - Lazy Learner – very simple “training” phase
 - *kNN* in scikit-learn: <https://scikit-learn.org/stable/modules/neighbors.html>
- ◆ Hyperparameters
- ◆ Curse-of-Dimensionality





Your Turn !

Exercise 3 (Homework)

kNN - coding

